EXP: 12 (a) Implementation of echo client server using TCP/IP sockets

Echo client Server using TCP Socket

AIM:

To implement an echo-client-server application using TCP sockets.

ALGORITHM:

1. Create a TCP Socket.
2. Bind the it to an IP address and port.
3. Listen for client connection.
4. accept the client connection.
5. recieve the message and send it Back
6. close the connection

CLIENT:

1. create a TCP socket
2. connect to the server
3. Send a message
4. recieve a echoed message.
5. Display the message.
6. close the connection

Code:

Server

```
import socket
s = socket.socket()
s = bind ("localhost", 5000)
s = listen(1)
c, addr = s.accept()
msg = c.recv(1024).decode()
c.send(msg.encode())
c.close()
s.close()
```

Input :

Enter message Hello server.

Output :

Server : Hello Server

CLIENT

```
import socket
S = socket.socket()
S.connect (("localhost", 5000))
msg = input ("Enter message")
S.send (msg.encode())
Print ("Echo:", S.recv(1024).decode())
S.close()
```

Result:

Thus the echo client server was successfully implemented using the TCP sockets

EXP 12 (b) Implementation of a chat program using socket programming

CHAT CLIENT SERVER USING TCP SOCKETS

AIM:

To implement a simple chat client - server application using TCP sockets.

ALGORITHM:

Server

1. create a TCP socket
2. Bind and listen for a client.
3. Accept the connection.
4. Recieve a message from the client.
5. Send a replay to the client.
6. Close the connection.

CLIENT - Server

```
import socket
S= socket.socket()
S.bind("localhost",5006)
S.listen(1)
c1 addr = s accept()
while True:
    msg = c.recv(1024).decode()
    if msg = "bye": break.
    Print("client", msg)
    c.send(input("Server").encode())
    c.close()
    s.close()
```

code. client.

```
import socket
S = socket.socket()
S = connect(("localhost", 5000))
while true:
    msg = input("client:")
    S.send(msg.encode())
    if msg = "bye": break.
```

print ("server", server (1024 . decode ())

S.close ()

Result :

Thus the chat client server was successfully implemented whij the TCP socket.